

Endogrid: Measuring the Cost of Reasoning

A long-horizon economic planning benchmark where building more can be worth less.

What it is

An agent gets a procedurally generated terrain, \$25M, and ten simulated years. It sites solar, wind and hydro, orients them, wires them to demand zones, finances them with cash and debt. Every simulated hour the physics resolves and a market clears. The objective is the **equity value** of the resulting business — not megawatt-hours, not cash.

Price is **endogenous**: it clears each tick against residual demand served by a thermal fleet in merit order. Renewables bid at zero marginal cost, so an agent’s own output displaces the marginal unit and lowers the price it earns — supply 0kW and power clears at \$79/MWh, supply 30 MW and it clears at −\$3. Each asset is discounted at its *own* realised capture price, so flooding the market marks down what you already own. The naive maximising policy defeats itself, and wealth can fall.

Representative research efforts in the area

	World	Objective	Horizon	Adaptation	Cost axis
Sudoku-Extreme [1]	fixed puzzle	correct grid	one shot	recurrent state	no
ARC-AGI-2 [5]	held-out tasks	correct grid	one shot	in-context	no
HRM [3]	puzzle suite	correct grid	one shot	backward pass	params only
TRM [4]	puzzle suite	correct grid	one shot	recursive	params only
Crafter [6]	procedural grid	achievements	episode	policy weights	no
MineDojo [7]	open world	task suite	episode	policy weights	no
Endogrid	procedural grid	equity value	10 sim-years	in-episode state	dollars

We borrow evaluation *logic* from the BDH line, not the model. BDH is judged on correct Sudoku solutions with no chain-of-thought and no backtracking (97.4% top-1) [1]; BDH-CQ adapts to unseen ARC tasks through recurrent state with no parameter updates [2]. Our agents get the same constraints: no weight updates during the ten-year run, no search, no backtracking — a built machine is a spent \$1.45M. HRM [3] is the contrast that makes “no fine-tuning mid-run” literature-grounded rather than invented, since the optimisation route demonstrably exists. Held-out seeds follow ARC-AGI-2’s private-set methodology [5]: seeds 1–8 development, 9–10 untouched until final scoring.

Results

Agent (10 seeds, 10 sim-years)	Score	sd	Equity	Return
heuristic — no search, no backtracking	43.8	11.5	\$46.5M	+86%
random	22.2	11.4	\$28.3M	+13%
donothing — holds capital	16.3	0.0	\$25.0M	0%
lookup — fixed terrain→machine table	9.1	2.5	\$17.0M	− 32%

Memorisation loses money. lookup scores below doing nothing, on every seed. An agent applying a fixed terrain→machine table does not merely underperform — it destroys a third of its capital, because it sites without regard to the resource that actually reaches a cell and orients every machine identically regardless of wind. This is the falsifiable form of the anti-causal-confusion design [8].

The expensive route buys little. Same map, same twenty-decision budget:

	Score	Cost	Wall-clock	Score/\$
Sonnet 5 (explicit reasoning)	43.4	\$0.80	3.8 min	54
heuristic (state only, no search)	40.8	\$0.0012	0.7 min	32,870

94% of the score at 1/670 of the cost — the TRM-versus-HRM parameter-count argument [4,3] (7M vs 27M vs BDH-CQ’s larger recurrent model) restated in dollars, on a ten-year horizon with an economic objective rather than one grid.

Limitations and unanswered questions

No BDH model was run. This is a harness, a protocol and a measurement. The high-score/low-cost corner is *empty* — nothing we ran occupies it. That corner is what BDH-style recurrent-state architectures claim, and this benchmark makes the claim testable in dollars. It is not evidence they win.

The causal-confusion ablation is unrun. We designed against spurious terrain shortcuts and show a lookup table loses money, but have *not* shown our reasoning agents are free of some *other* shortcut [8]. The honest test permutes the cosmetic identity of terrain types with physics held fixed; an agent using a genuine causal chain should be unaffected. Designed, not run — the first thing we would do next.

Language-model results are single-seed, on an engine since corrected for a defect those very episodes exposed: an agent reported hydro sites showing zero head and was right — the validator checked flow but never head. They calibrate cost; they do not rank models.

Memory here is environmental, not a cache. Soil moisture, lagged discharge and a seasonally mean-reverting wind bearing persist whether or not the agent attends to them — closer to BDH’s Hebbian fast-weight memory [1] than to a key–value cache. Whether an agent must *internalise* that state or may re-read it each tick is open.

Reproducing: deterministic — seed + config + action log give a bit-identical state hash. 41 executable invariants; interface in docs/API.md.

References. [1] Kosowski et al. *The Dragon Hatchling*. arXiv:2509.26507, 2025. [2] Engdahl et al. *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888, 2026. [3] Wang et al. *Hierarchical Reasoning Model*. arXiv:2506.21734, 2025. [4] Jolicoeur-Martineau. *Less is More: Recursive Reasoning with Tiny Networks*. arXiv:2510.04871, 2025. [5] Chollet et al. *ARC-AGI-2*. arXiv:2505.11831, 2025. [6] Hafner. *Benchmarking the Spectrum of Agent Capabilities*. ICLR 2022. [7] Fan et al. *MineDojo*. NeurIPS 2022. [8] Tien et al. *Causal Confusion and Reward Misidentification*. ICLR 2023.